

Indian Brain Behind New Painless Cancer Detection Device



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Dr Srirang Manohar is the man who gave PAM a fillip

According to World Cancer Research Fund International, breast cancer is the most common type of cancer in women, and the second most common cancer overall. Doctors recommend that women frequently take tests to detect the onset of cancer early, when it can still be treated. Yet, current procedures for cancer detection put many women off, especially in rural areas.

A new method, proposed by a team from University of Twente, The Netherlands, in *Nature Scientific Reports* (refer www.nature.com/articles/srep11778#ref48) could offer a way out. The paper presents a photoacoustic method for imaging soft tissues, which has, in turn, been applied to develop a non-invasive, economical and effective method to detect breast cancer.

We caught up with Dr Srirang Manohar, associate professor at University of

Twente's Biomedical Photonic Imaging group to find out more about this new technology, which they call PAMmography.

Concept to device

In 2001, Dr Manohar, who did his PhD from Indian Institute of Science (IISc), Bengaluru, learnt of a vacancy at University of Twente, where some pioneering work was being conducted in photoacoustics, involving light interactions with biological tissues. In the new project, applicability of the method to breast imaging was to be investigated.

Dr Manohar's involvement gave the project a fillip, and four years later they had a working prototype called Twente photoacoustic mammoscope (PAM). The device has consistently evolved since then, leading to the recent technical paper, which reports a successful study on 29 patients using an upgraded ver-

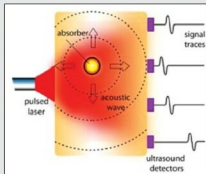
Imaging with light

The key technology behind PAMmography is photoacoustic imaging. Methodology:

1. Light pulses are applied to the tissue as a probing energy beam, with the aim to visualise sites where optical absorption takes place in the tissue.
2. Absorbed light is converted into heat, which causes a rise in temperature, resulting in thermal expansion.
3. When short pulses of light are used, thermal expansion causes mechanical waves to be generated from areas where light was absorbed.
4. The mechanical wave has frequency components in the ultrasound regime and can be detected at the boundary of the tissue using ultrasound detectors.
5. From this point on, common ultrasound imaging takes over.

The use of light for imaging soft tissues provides many advantages:

1. Interaction between light and haemoglobin gives a high absorption contrast, which results in a good-quality image.
2. Red and near-infrared wavelengths used by this method constitute non-ionising radiation, which means the tissue is unaffected by tests.
3. It also enables spectroscopy to identify more physiological information about the tissue.
4. Use of light ensures that this technique is non-invasive. Patients just have to lie, breast down, on the mammoscope, for the tests to be conducted.



The concept behind PAM